

ViLo: LiDAR Localization with **V**ision-**L**anguage Priors

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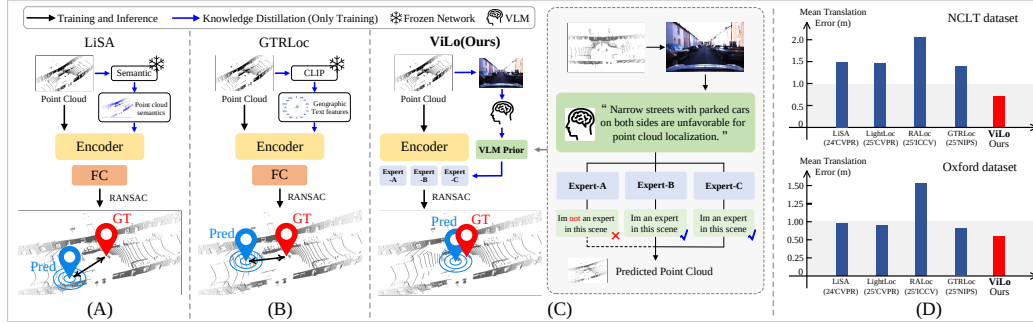


Figure 1: (A) LiSA and (B) GTRLoc rely on a single network to learn features indiscriminately, lacking scene-adaptive capability. (C) The proposed ViLo leverages VLM priors to guide multiple expert networks for more scene-adaptive learning. (D) Comparison of performance with current state-of-the-art methods on the Oxford and NCLT outdoor LiDAR localization benchmarks.

Abstract

LiDAR localization is a fundamental task in robotics and computer vision, aiming to estimate the global pose of point clouds. Although Scene Coordinate Regression (SCR) has demonstrated state-of-the-art performance in this field, standard SCR methods employ a monolithic network for uniform optimization across diverse scenes, which inevitably suffers from capacity interference in highly degraded scenes. Recent research on Vision-Language Models (VLMs) indicates that they can acquire rich scene understanding priors, providing the adaptive learning capabilities currently missing in localization networks. In this paper, we propose ViLo, the first framework to integrate VLM priors into SCR to fundamentally enhance localization robustness. Specifically, we design a temporally consistent key-frame querying mechanism to extract open-vocabulary degeneracy priors. We then construct a dynamic prior-guided Mixture-of-Experts (MoE) model for adaptive feature learning tailored to varying scenes. Finally, to avoid additional computational overhead during inference, ViLo internalizes the VLM’s scene understanding into the 3D backbone via cross-modal distillation, enabling a lightweight, LiDAR-only deployment. Extensive experiments on the Oxford RobotCar and NCLT datasets demonstrate that ViLo significantly outperforms existing state-of-the-art methods, reducing positional errors by impressive margins of 20% and 51%, respectively.

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1 Introduction

LiDAR localization, which aims to estimate the 6-DoF global pose of point clouds, is a fundamental and critical task in robotics and autonomous driving [1, 2, 3, 4]. Traditional map-based methods [5, 6] rely on dense matching between query frames and pre-built 3D maps, which often incurs prohibitive storage costs and high communication overhead [7, 8, 9]. To overcome these limitations, learning-based map-free regression methods have emerged [10, 11, 12, 13]. These methods are generally categorized into Absolute Pose Regression (APR) and Scene Coordinate Regression (SCR). Unlike APR [14, 15], which directly regresses poses, SCR [16] predicts correspondences between the point cloud and world coordinates and subsequently solves for the pose via RANSAC [17]. Due to its explicit exploitation of geometric information, SCR has demonstrated superior performance and yielded numerous state-of-the-art results in large-scale outdoor environments [18]. For instance, LiSA [19] enhances performance by distilling semantic features into the SCR network to help the model understand deep point cloud semantics. GTR-Loc [20] leverages CLIP [21] to obtain geospatial text features, aiding the network in reducing the impact of visually similar geometric features. Additionally, RALoc [18] utilizes data augmentation and feature fusion to mitigate the influence of environmental structural changes.

Despite these impressive results, as shown in Fig. 1, standard SCR methods [12, 18, 19, 22, 20] face a fundamental bottleneck in large-scale outdoor scenarios: they typically employ a monolithic network for uniform optimization across significantly diverse environments [23, 24]. Real-world outdoor settings exhibit extreme structural diversity, ranging from "nominal environments" rich in structural features to "highly degraded scenes" characterized by severe occlusions, dynamic objects, or geometric homogeneity. Forcing a monolithic network to perform uniform fitting across such divergent patterns inevitably leads to severe Capacity Interference, where the network struggles to find an effective balance between reliable geometric structures and environmental noise.

To fundamentally break through this limitation, we propose ViLo, the first SCR localization framework to deeply integrate high-level Vision-Language Model (VLM) priors with a Mixture-of-Experts (MoE) architecture [25]. Specifically, we first introduce an efficient key-frame querying mechanism to obtain temporally consistent, open-vocabulary [26] VLM degeneracy priors. Subsequently, we construct a Scene-Adaptive MoE model. By combining hierarchical scale-aware experts with scene routing and adaptive gating mechanisms, ViLo dynamically schedules the expert network based on VLM scene priors. This approach effectively eliminates the capacity interference issues inherent in monolithic networks, enabling robust localization in complex and dynamic scenarios.

Finally, considering the prohibitive computational latency of invoking large VLMs or processing multi-modal images during real-time inference, ViLo adopts a Cross-modal Distillation strategy [27, 28]. During training, the VLM's scene understanding and adaptive scheduling capabilities are internalized into a pure 3D LiDAR backbone. At inference time, the model entirely discards image inputs and the VLM, enabling a lightweight, real-time, zero-overhead, LiDAR-only deployment.

We evaluate ViLo on two large-scale outdoor benchmarks, including Oxford RobotCar [29] and NCLT [30]. Results demonstrate that ViLo significantly outperforms existing top-tier methods. Our primary contributions are summarized as follows:

- We propose ViLo, the first LiDAR localization framework to integrate VLM priors into SCR, fundamentally enhancing robustness and addressing scene-level capacity interference.
- We design an efficient key-frame querying mechanism to obtain stable, open-vocabulary degeneracy priors from the VLM, providing crucial guidance for adaptive optimization.
- We introduce a Scene-adaptive MOE that dynamically adapts to diverse environments via scene routing and adaptive gating, overcoming the limitations of monolithic uniform optimization.
- We establish a cross-modal distillation strategy to internalize VLM reasoning into the 3D backbone, enabling a real-time, LiDAR-only inference pipeline that surpasses state-of-the-art methods by 20% and 51% on Oxford and NCLT.

2 Related Work

LiDAR Localization. Traditional LiDAR localization methods typically aim to establish correspondences between query point clouds and pre-built 3D maps [2, 31, 32]. These methods primarily

achieve localization through map matching. Although map matching can provide high accuracy, it suffers from prohibitive map storage costs and significant communication overhead. To address these limitations, researchers have proposed regression-based localization methods, which perform inference without relying on pre-built 3D maps.

Absolute Pose Regression (APR). Absolute Pose Regression methods [13, 15] directly regress 6-DoF poses from input point clouds. Pioneering research such as PointLoc [10] utilized PointNet++ [33] with a multi-layer perceptron (MLP) head to validate the feasibility of this approach, though it showed limitations in complex scenes. Subsequent studies have improved APR through novel network architectures, multi-modal fusion, or temporal constraints [14]. For example, DiffLoc [15] optimizes poses via an iterative diffusion process. Nevertheless, APR’s reliance on global scene representations can hinder effective geometric encoding, which potentially limits localization accuracy.

Scene Coordinate Regression (SCR). Unlike APR, Scene Coordinate Regression methods learn to predict per-point world coordinates and determine the final pose through RANSAC [17]. SGLoc [12] first decoupled the localization task into point correspondence regression and pose estimation. To enhance training efficiency in large-scale scenes, LightLoc [22] constructs a pre-trained encoder that achieves optimal performance with just one hour of training. PELoc [16] utilizes data augmentation to achieve compatibility between low-cost rapid deployment and localization accuracy. Although SCR methods achieve higher accuracy by learning scene geometry, they perform poorly in highly degraded or visually similar scenes. To mitigate this issue, semantic-aware localization LiSA [19] distills a semantic segmentation model to help the network understand point cloud structures. The geometric-text-aware method GTRLoc [20] utilizes CLIP’s pre-trained text features [21] to reduce feature ambiguity. RALoc [18] employs rotation augmentation and feature fusion strategies to reduce the impact of environmental structural noise. However, these methods primarily focus on feature-level enhancement and fundamentally rely on a monolithic network for uniform optimization across diverse scenes, which inevitably leads to capacity interference.

Vision-Language Models (VLM). Benefiting from the development of Large Language Models (LLMs) such as ChatGPT [34], LLaMA [35], and Gemini [36], the use of universal priors has become possible, stimulating research in the Vision-Language Model (VLM) field. Models like CLIP [21] and VL-Mamba [37] use large-scale pre-training to align textual and visual representations, facilitating the development of VLM-based localization tasks [38]. For instance, LP-SAM [39] and TextSLAM [40] first achieved text-based place recognition, while Text2Pos [41] and Text2Loc [42] proposed using text for large-scale urban place recognition. The latest MSGNav [43] even enables training-free, zero-shot indoor navigation based on VLMs. However, the aforementioned methods use VLMs for feature matching or indoor navigation but lack the capability for adaptive architectural scheduling in diverse outdoor scenes. To this end, we propose ViLo, which leverages VLMs to obtain scene priors to dynamically guide adaptive localization for different experts across various scenes. Through a distillation strategy, VLM reasoning capabilities are internalized into the 3D backbone, enabling LiDAR-only localization without additional overhead during inference.

3 Methodology

ViLo relies on Scene Coordinate Regression (SCR) for LiDAR localization. The input is a query point cloud $\mathcal{P} = \{p_i\}_{i=1}^N$, where $p_i \in \mathbb{R}^3$ represents a 3D point in the local LiDAR coordinate frame. The output is a set of predicted world coordinates $\mathcal{P}' = \{p'_i\}_{i=1}^N$. The standard training loss for SCR is typically defined as L_1 distance between the predicted p'_i and the ground-truth p_i^* :

$$\mathcal{L}_{\text{loc}} = \frac{1}{N} \sum_{i=1}^N \|p'_i - p_i^*\|_1. \quad (1)$$

3.1 Motivation

The core idea of ViLo is to leverage VLM priors to assist the localization task. To illustrate our rationale, Fig. 2 presents: (A) the advantage of VLM reasoning over rigid semantics, and (B) the necessity of the MoE architecture for diverse scenes. As shown in Fig. 2(A), compared to the semantic-aware method LiSA [19], which misidentifies dynamic buses as stable landmarks and forces the network to learn unreliable geometries, ViLo transfers 2D scene priors to the 3D backbone. This enables the network to perceive the localizability of the current environment and autonomously schedule feature learning for improved performance. Furthermore, Fig. 2(B) highlights the limitations

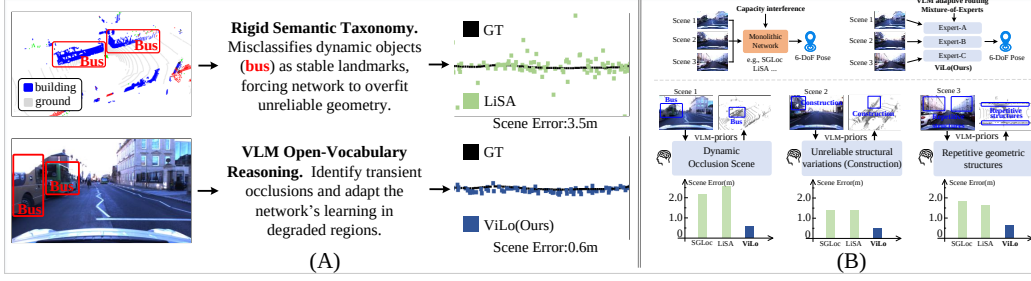


Figure 2: **Motivation.** Left(A): VLM priors are adopted to better perceive scene localizability compared to semantic-aware methods (LiSA). Right(B): Mixture-of-Experts (MOE) is utilized to resolve the capacity interference inherent in monolithic networks across various degraded scenes through VLM-guided dynamic expert scheduling.

of monolithic architectures. While existing methods like SGLoc [12] and LiSA [19] rely on feature-level fusion within a single network, they suffer from severe capacity interference during uniform optimization. In contrast, ViLo utilizes VLM priors to guide specialized learning across distinct expert networks, effectively resolving these conflicts. To this end, as shown in Fig. 3, we propose ViLo, a novel framework designed to efficiently integrate VLM priors into SCR-based localization.

3.2 VLM Priors

To ensure computational efficiency and stability, we first propose a temporally consistent key-frame querying mechanism. Subsequently, we leverage the VLM’s Chain-of-Thought [44, 45] capability to explicitly reason about scene degeneracies from 2D images. Finally, we detail how these raw priors are extracted into optimization weights and scene semantic embeddings for downstream tasks.

Key-Frame Querying and Temporal Consistency Independent per-frame evaluation lacks temporal consistency and incurs prohibitive API costs. To balance evaluation stability and efficiency, we introduce a temporal consistency prompting mechanism by replacing exhaustive querying with spatial key-frames. A key-frame, denoted as $\mathcal{K}_i$, is selected when either of the following criteria is met: (1) the spatial displacement reaches the LiDAR sparse Field of View boundary, or (2) the visual similarity threshold between consecutive frames are satisfied (see **Appendix Sec. A**). When evaluating the current key-frame $\mathcal{K}_i$, we append the VLM results of the previous N key-frames ($\mathcal{K}_{i-1}$ to $\mathcal{K}_{i-N}$), which include the degradation score s and the degeneracy description $\mathcal{T}^{\text{desc}}$. This historical context enforces reasoning smoothness and suppresses evaluation fluctuations across continuous scenes. Intermediate non-key-frames efficiently inherit these priors via interpolation. This strategy reduces token overhead by $\sim 95\%$ while preserving robust scene-level guidance.

Open-Vocabulary Degeneracy Reasoning Traditional semantic methods rely on rigid classification taxonomies, failing to capture the complexity of real-world environments. We address this by guiding the VLM through a 2D-to-3D Chain-of-Thought to analyze outdoor structures from first principles. For a given key-frame $\mathcal{K}_i$, the VLM generates free-form, open-vocabulary outputs comprising three essential priors: (1) a localizability score $s_i \in [0, 10]$, where higher values indicate more severe degradation; (2) a degeneracy description $\mathcal{T}_i^{\text{desc}}$ for temporal prompting, such as "dynamic bus occlusion", "seasonal foliage variations", and "building construction"; and (3) a scene content analysis $\mathcal{T}_i^{\text{content}}$. Prompt details in **Appendix Sec. B**.

VLM Prior Extraction The VLM priors provide two key types of guidance for point cloud relocalization, as follows:

1) Global Optimization Weight (α_i): The normalized evaluation score s_i serves as a global weight to modulate the overall training loss. It acts as a scene-adaptive curriculum learning strategy, preventing network gradients from being dominated by structural noise in highly degraded environments.

2) Scene Semantic Embedding (t_i): We utilize a frozen, lightweight text encoder $M(\cdot)$ to extract a semantic embedding $t_i = M(\mathcal{T}_i^{\text{content}})$ from the scene content analysis $\mathcal{T}_i^{\text{content}}$. This embedding provides the scene context necessary to guide the expert routing in MoE (Sec. 3.3)

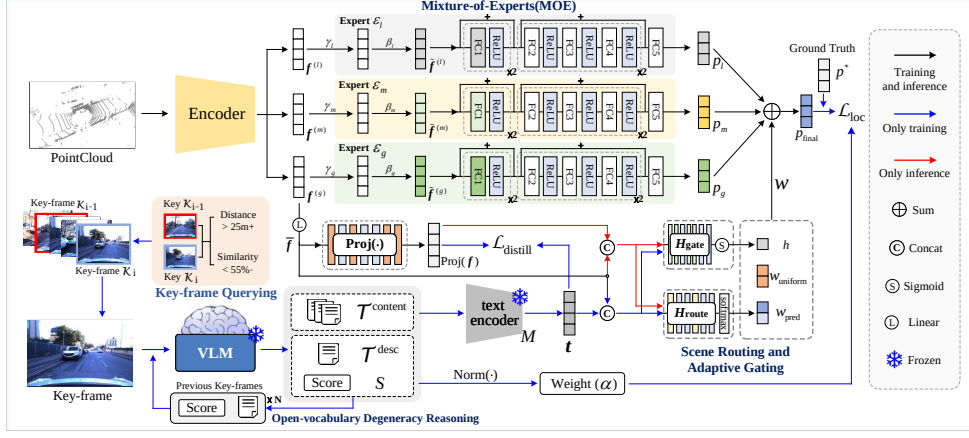


Figure 3: **The pipeline of ViLo.** We extract VLM priors via temporally consistent Key-frame Querying coupled with Open-vocabulary Degeneracy Reasoning. These priors are encoded by a frozen text encoder to extract deep semantic embeddings, which subsequently guide the Mixture-of-Experts via Scene Routing and Adaptive Gating to predict point clouds in the world coordinate system. Finally, the network is optimized by a localization loss ($\mathcal{L}_{loc}$) modulated by the prior information, alongside a cross-modal distillation loss ($\mathcal{L}_{distill}$) derived from the VLM’s scene reasoning. Finally, LiDAR-only inference enables lightweight deployment.

3.3 Scene-Adaptive Mixture-of-Experts with Scene Routing

To overcome monolithic capacity interference, we introduce a Scene-Adaptive Mixture-of-Experts (MoE). It leverages VLM priors to dynamically route and gate scale-aware experts, tailoring optimization across varied scenes without conflicting geometric fitting.

Hierarchical Scale-Aware Tri-Experts To prevent expert homogenization, We deploy three parameter-independent experts ($\mathcal{E}_l, \mathcal{E}_m, \mathcal{E}_g$) within a residual architecture, each geometrically grounded at different depths of the 3D backbone [22]. Extracting features from distinct layers yields varying receptive fields to capture specific geometries: high-frequency local details $\mathbf{f}^{(l)}$, mid-range regional structures $\mathbf{f}^{(m)}$, and global contextual topologies $\mathbf{f}^{(g)}$. Furthermore, to explicitly amplify feature discrepancy and ensure specialization, we introduce learnable scaling (γ_k) and shifting (β_k) factors for each expert $k \in \{l, m, g\}$. The backbone feature $\mathbf{f}^{(k)}$ is expressed as:

$$\tilde{\mathbf{f}}^{(k)} = \gamma_k \odot \mathbf{f}^{(k)} + \beta_k, \quad (2)$$

where $\odot$ denotes element-wise multiplication. This mechanism enforces representation divergence, ensuring scale-specific specialization. Each expert then processes the modulated feature to output coordinate predictions $p_k = \mathcal{E}_k(\tilde{\mathbf{f}}^{(k)})$.

Scene Routing and Adaptive Gating Determining the optimal expert requires holistic awareness of both the environmental semantics and the current geometric layout. Thus, instead of relying solely on the textual prior, we concatenate the global point cloud feature $\bar{\mathbf{f}}_i = \text{Linear}(\mathbf{f}_i^{(g)})$ with the VLM semantic embedding $\mathbf{t}_i$ to form a cross-modal context vector $\mathbf{v}_i = [\bar{\mathbf{f}}_i, \mathbf{t}_i]$. A routing network $H_{\text{route}}(\cdot)$ then projects this fused context into categorical probabilities across the three experts:

$$\mathbf{w}_{\text{pred}} = \text{Softmax}(H_{\text{route}}(\mathbf{v}_i)). \quad (3)$$

Although the cross-modal context provides robust guidance, VLM features can occasionally be noisy or misaligned with the geometric reality. To guarantee training stability, we introduce an Adaptive Gating Mechanism. We train a confidence gate $h_i = \sigma(H_{\text{gate}}(\mathbf{v}_i)) \in [0, 1]$, where σ denotes the Sigmoid function. The final routing weight vector $\mathbf{w}_i = [w_l, w_m, w_g]$ interpolates between the dynamically predicted routing and an unbiased uniform prior:

$$\mathbf{w}_i = h_i \cdot \mathbf{w}_{\text{pred}} + (1 - h_i) \cdot \mathbf{w}_{\text{uniform}}, \quad (4)$$

where w_{uniform} denotes a uniform weight. When the cross-modal context is highly confident ($h_i \rightarrow 1$), the network strictly follows the specialized routing. Conversely, when the scene semantics are ambiguous ($h_i \rightarrow 0$), the routing gracefully degrades to a uniform distribution, preventing expert collapse. The final coordinate prediction is aggregated as:

$$p_{\text{final}} = \sum_{k \in \{l, m, g\}} w_k \cdot p_k. \quad (5)$$

The aggregated p_{final} is then used as p' in Eq. 1 to compute the localization loss $\mathcal{L}_{\text{loc}}$.

Cross-Modal Scene Semantic Distillation To internalize the VLM’s scene reasoning into the 3D backbone, we project the global LiDAR feature $\mathbf{f}_i$ into the text-aligned latent space via a projection head $\text{Proj}(\cdot)$. A distillation loss $\mathcal{L}_{\text{distill}}$, based on cosine similarity $\cos(\cdot)$, aligns the implicit features $\text{Proj}(\mathbf{f}_i)$ with VLM scene semantic embedding $\mathbf{t}_i$:

$$\mathcal{L}_{\text{distill}} = 1 - \cos(\text{Proj}(\mathbf{f}_i), \mathbf{t}_i). \quad (6)$$

3.4 Loss Function and LiDAR-only Inference

Training Loss The total training objective is defined as:

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{loc}} + \lambda \mathcal{L}_{\text{distill}}, \quad (7)$$

where α denotes the global optimization weight derived from the VLM priors. It modulates the overall loss to prevent network gradients from being dominated by structural noise in highly degraded environments, and λ is a balancing parameter empirically set to 0.1.

LiDAR-Only Inference During inference, ViLo operates strictly on LiDAR inputs. Since the VLM priors are successfully distilled into the 3D backbone via $\mathcal{L}_{\text{distill}}$, the projected global LiDAR feature $\text{Proj}(\mathbf{f}_i)$ serves as an effective surrogate for the scene semantic embedding $\mathbf{t}_i$. Thus, at test time, the context vector is constructed exclusively from the point cloud as $\hat{\mathbf{v}}_i = [\mathbf{f}_i, \text{Proj}(\mathbf{f}_i)]$, from which both the scene routing probabilities and the adaptive gate are directly predicted:

$$\mathbf{w}_{\text{pred}} = \text{Softmax}(H_{\text{route}}(\hat{\mathbf{v}}_i)), \quad (8)$$

$$h_{\text{pred}} = \sigma(H_{\text{gate}}(\hat{\mathbf{v}}_i)). \quad (9)$$

The final routing weight $\mathbf{w}$ is then computed by interpolating $\mathbf{w}_{\text{pred}}$ with the uniform prior $\mathbf{w}_{\text{uniform}}$ using h_{pred} , exactly as formulated in Eq. 4.

4 Experiments

4.1 Setup

Datasets and Evaluation Metrics We evaluate on two outdoor LiDAR localization benchmarks: Quality-enhanced Oxford RobotCar [29] and NCLT [30], using mean position error (m) and mean orientation error ($^\circ$) as metrics. The datasets are detailed as follows:

Quality-enhanced Oxford RobotCar: This urban driving dataset covers a 10 km route under various weather conditions, processed using the ground truth optimization method from SGLoc [12]. We use sequences 11-14-02-26, 14-12-05-52, 14-14-48-55, and 18-15-20-12 for training, and sequences 15-13-06-37, 17-13-26-39, 17-14-03-00, and 18-14-14-42 for testing.

NCLT: Collected at the University of Michigan North Campus, this dataset spans a 5.5 km route and captures seasonal variations. Ground truth is obtained by fusing GPS with SLAM optimization. Following prior localization settings [12, 19, 22], we use 2012-01-22, 2012-02-02, 2012-02-18, and 2012-05-11 for training, and 2012-02-12, 2012-02-19, 2012-03-31, and 2012-05-26 for testing.

Baselines We compare with state-of-the-art large-scale LiDAR localization methods, including absolute pose regression (APR) methods PosePN++ [11], STCLoc [14], and HypLiLoc [13], as well as scene coordinate regression (SCR) methods SGLoc [12], LiSA [19], LightLoc [22], RALoc [18], and GTRLLoc [20]. Among them, LightLoc, RALoc, and GTRLLoc are the most recent methods.

Table 1: Mean position and orientation error $\downarrow$ (m, $^\circ$) on the Quality-enhanced Oxford dataset. We highlight the **best** and **second-best** results.

Method		15-13-06-37	17-13-26-39	17-14-03-00	18-14-14-42	Mean[m, $^\circ$] $\downarrow$
APR	PosePN++ [11] (2022'PR)	4.54, 1.83	6.44, 1.78	4.89, 1.55	4.64, 1.61	5.13, 1.69
	STCLoc [14] (2022'TITS)	5.14, 1.27	6.12, 1.21	5.32, 1.08	4.76, 1.19	5.34, 1.19
	HypLiLoc [13] (2023'CVPR)	5.03, 1.46	4.31, 1.43	3.61, 1.11	2.61, 1.09	3.89, 1.27
SCR	SGLoc [12] (2023'CVPR)	1.79, 1.67	1.81, 1.76	1.33, 1.59	1.19, 1.39	1.53, 1.60
	LiSA [19] (2024'CVPR)	0.94, 1.10	1.17, 1.21	0.84, 1.15	0.85, 1.11	0.95, 1.14
	LightLoc [22] (2025'CVPR)	0.82, 1.12	0.85, 1.07	0.81, 1.11	0.82, 1.16	0.83, 1.12
	RALoc [18] (2025'ICCV)	1.52, 1.28	1.71, 1.27	1.37, 1.20	1.41, 1.23	1.51, 1.24
	GTRLoc [20] (2025'NeurIPS)	<u>0.77, 1.02</u>	<u>0.77, 1.01</u>	<u>0.67, 1.01</u>	<u>0.80, 1.07</u>	<u>0.75, 1.03</u>
	ViLo(Ours)	0.65, 0.87	0.60, 0.82	0.53, 0.83	0.61, 0.92	0.60, 0.86

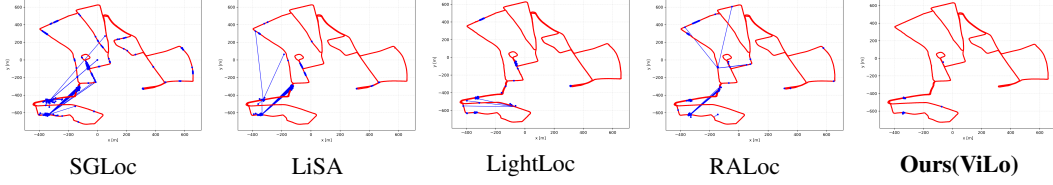


Figure 4: Visualization on the Quality-enhanced Oxford (Seq. 17-13-26-39). Red and black points denote predictions and ground truth, respectively, localization errors >5 m are shown in blue points.

Implementation ViLo is implemented in PyTorch with MinkowskiEngine for sparse convolutions. Voxel sizes are set to 0.25 m on Oxford and 0.3 m on NCLT. We use the AdamW optimizer [46] with a One-cycle learning rate schedule from $5e-4$ to $5e-3$, a batch size of 256, and train for 150 epochs. The VLM calls APIs of Gemini 2.5 pro [36], while a frozen Qwen3-Embedding-0.6B [47] extracts 256-dimensional VLM scene text features. All experiments are conducted on a single NVIDIA RTX 4090 GPU. More training details and network architecture ($H_{route}(\cdot)$, $H_{gate}(\cdot)$, $Proj(\cdot)$) implementation can be found in **Appendix Sec. C**.

4.2 Results on Oxford Dataset

Tab. 1 summarizes the localization performance comparison. Our proposed ViLo achieves state-of-the-art results in both translation and rotation accuracy. Compared to the second-best method GTRLoc [20], ViLo improves the average localization performance by 20% (from 0.75 m to 0.60 m) and the average orientation performance by 17% (from 1.03° to 0.86°). As shown in Fig. 4, while baseline methods frequently suffer from catastrophic localization failures (blue points: large deviations from ground truth), ViLo maintains robust performance with significantly fewer outliers.

4.3 Results on NCLT Dataset

Tab. 2 presents the results on the NCLT benchmark. ViLo achieves an average localization error of 0.69 m and an orientation error of 2.28° . Compared to the best baseline GTRLoc, ViLo reduces the localization error by 51% (from 1.40 m to 0.69 m). To our knowledge, ViLo is the first method to achieve sub-meter LiDAR localization on the NCLT dataset. As shown in Fig. 5, while all baseline methods produce severe localization errors in challenging regions, ViLo achieves robust performance with almost no predictions exceeding 5 m error.

4.4 Runtime and Model Size

Inference time is critical for localization. With LiDAR running at 20 Hz on Oxford and 10 Hz on NCLT, real-time inference requires under 50 ms and 100 ms per frame, respectively. As shown in Tab. 3, Although VLM priors are used in training via distillation, ViLo needs only point cloud input to the SCR module at inference. It runs at 33 ms (<50 ms) on Oxford and 53 ms (<100 ms) on NCLT, meeting real-time demands for autonomous driving.

Table 3: Runtime and Model Size Comparison on Oxford and NCLT.

Method	Model Size $\downarrow$	Runtime $\downarrow$	
		Oxford(<50 ms)	NCLT(<100 ms)
SGLoc [12]	116MB	38ms	75ms
LiSA [19]	116MB	38ms	75ms
LightLoc [22]	74MB	29ms	48ms
RALoc [18]	123MB	26ms	55ms
ViLo(Ours)	92MB	33ms	53ms

Table 2: Mean position and orientation error $\downarrow$ (m, $^\circ$) on the NCLT dataset. We highlight the **best** and **second-best** results.

Method		2012-02-12	2012-02-19	2012-03-31	2012-05-26	Mean[m, $^\circ$] $\downarrow$
APR	PosePN++ [11] (2022'PR)	4.97, 3.75	3.68, 2.65	4.35, 3.38	8.42, 4.30	5.36, 3.52
	STCLoc [14] (2022'TITS)	4.91, 4.34	3.25, 3.10	3.75, 4.04	7.53, 4.95	4.86, 4.11
	HypLiLoc [13] (2023'CVPR)	1.71, 3.56	1.68, 2.69	1.52, 2.90	2.29, 3.34	1.80, 3.12
SCR	SGLoc [12] (2023'CVPR)	1.20, 3.08	1.20, 3.05	1.12, 3.28	3.48, 4.43	1.75, 3.46
	LiSA [19] (2024'CVPR)	0.97, 2.23	0.91, 2.09	0.87, 2.21	3.11, <u>2.72</u>	1.47, <u>2.31</u>
	LightLoc [22] (2025'CVPR)	0.98, 2.76	0.89, 2.51	0.86, 2.67	3.10, 3.26	1.46, 2.80
	RALoc [18] (2025'ICCV)	1.61, 4.71	1.61, 4.88	1.58, 4.71	3.51, 5.40	2.07, 4.92
	GTRLoc [20] (2025'NeurIPS)	<u>0.95</u> , 2.53	<u>0.82</u> , 2.45	<u>0.82</u> , 2.52	<u>3.01</u> , 2.99	<u>1.40</u> , 2.62
	ViLo(Ours)	0.64 , <u>2.39</u>	0.55 , <u>2.15</u>	0.59 , <u>2.22</u>	0.99 , 2.37	0.69 , 2.28

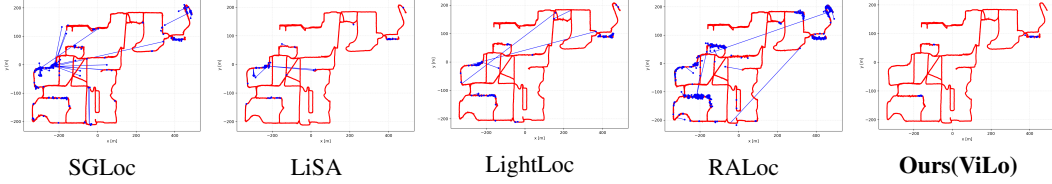


Figure 5: Visualization on the NCLT dataset (Seq. 2012-02-19). Red and black points denote predictions and ground truth, respectively, localization errors >5 m are shown in blue points.

Table 4: Ablation of Different VLM Priors on Oxford (w/o:without, w.:with).

Method	Mean Error $\downarrow$	
	Position(m)	Orientation($^\circ$)
ViLo (w/o VLM)	0.91	1.14
ViLo (w. Qwen2.5-7b)	0.93	1.17
ViLo (w. GPT-4o mini)	0.88	1.05
ViLo (w. Qwen 3.5 plus)	0.66	0.88
ViLo (w. Gemini 2.5 pro)	0.62	0.84

Table 5: Ablation on Scene Semantic Encoders (w/o:without, w.:with).

Method	Model Params	Position Error $\downarrow$	
		Oxford	NCLT
ViLo (w. Nomic)	0.15B	0.85	1.05
ViLo (w. BGE-M3)	0.5B	0.73	0.72
ViLo (w. Emb-gemma)	0.3B	0.66	0.74
ViLo (w. CLIP)	0.12B	0.77	0.85
ViLo (w. Qwen3-Emb)	0.6B	0.62	0.69

4.5 Ablation studies

4.5.1 Ablation on VLM Priors

Tab. 4 compares different VLM priors. Advanced models with robust scene reasoning, namely Gemini 2.5 Pro (default) [36] and Qwen 3.5 Plus [47], effectively guide the MoE, improving position accuracy by over 31% compared to the baseline without VLM. Conversely, lightweight VLMs like GPT-4o mini [48] and Qwen2.5-7b [49] lack the complex spatial reasoning capacity required for 2D-to-3D scene understanding, resulting in marginal gains or even slight performance degradation.

4.5.2 Ablation on Scene Semantic Encoders

Tab. 5 evaluates different scene semantic encoders for VLM priors: Qwen3-Embedding-0.6B (Qwen3-Emb) [47], CLIP [21], embeddinggemma(Emb-gemma) [50], BGE-M3 [51], and Nomic [52]. Accuracy consistently scales with model params. Our default Qwen3-Emb achieving the lowest errors overall. At smaller scales, the 0.12B CLIP outperforms the 0.15B Nomic due to explicit vision-language alignment. Notably, gemma-300m offers the best parameter-accuracy trade-off, outperforming the larger 0.5B BGE-M3 on Oxford and closely rivaling Qwen3-Emb on NCLT.

4.5.3 Ablation on ViLo Main Components

Tab. 6 ablates ViLo’s key components: Global Optimization Weight α (VLM-w), Scene Semantic Embedding t (VLM-t), and Mixture-of-Experts (MOE). While individual modules yield moderate gains, coupling VLM-t with MOE triggers a substantial performance leap, reducing the NCLT position error by 31.5% compared to the MOE-only variant. This underscores the necessity of semantic priors for expert routing. The full system achieves optimal precision, demonstrating strong synergy between global weighting and semantic-guided routing.

4.5.4 Ablation on the Number of Experts

To determine the optimal capacity of our Scene-Adaptive MoE, we ablate the expert count (E) in Table 7. The monolithic baseline ($E = \text{None}$) suffers from severe capacity interference, yielding

Table 6: Ablation of ViLo’s main components on Oxford and NCLT.

VLM-w	VLM-t	MOE	Mean Error [m, °] ↓	
			Oxford	NCLT
✓		✓	0.93, 1.15	1.51, 2.85
			0.85, 1.13	1.34, 2.66
			0.80, 1.02	1.27, 2.60
			0.67, 0.88	0.87, 2.47
✓	✓	✓	0.60, 0.86	0.69, 2.28

Table 7: Ablation on the Number of Experts.

Number of Expert	Mean Error [m, °] ↓	
	Oxford	NCLT
ViLo-(None)	0.93, 1.15	1.51, 2.85
ViLo-(2 Expert)	0.68, 0.90	0.98, 2.64
ViLo-(3 Expert)	0.60, 0.86	0.64, 2.28
ViLo-(4 Expert)	0.60, 0.83	0.67, 2.18

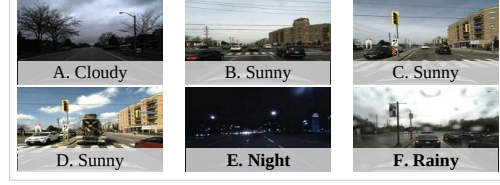


Figure 6: Boreas Dataset Scenes. We train on (A), (B), and (C), inference on (D), (E) and (F).

Table 8: ViLo LiDAR-only inference on Boreas Dataset.

Test Seq.	Different Conditions		
	D(Sun)	E(Night)	F(Rain)
Mean Error[m,°] ↓	2.33, 2.68	2.63, 2.48	2.71, 2.46

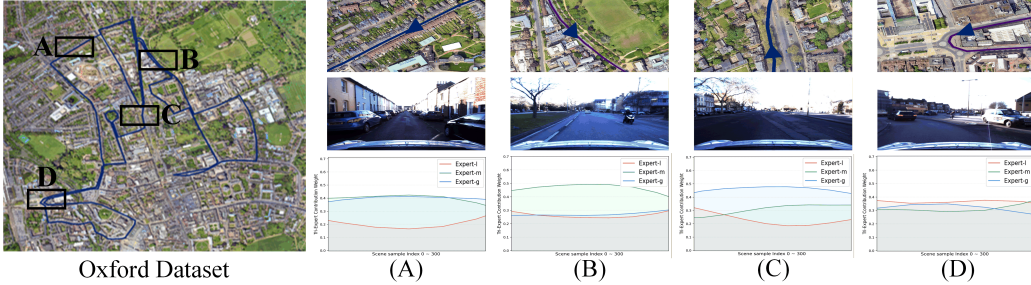


Figure 7: Visualization of MoE expert contribution weights across representative regions, demonstrating adaptive routing conditioned on VLM priors.

the highest errors. Introducing MoE routing drastically improves accuracy, with $E = 3$ achieving the optimal sweet spot (reducing NCLT position error to 0.64 m). Further scaling to $E = 4$ yields diminishing returns and slight degradation. This saturation is attributed to data dilution, where excessive experts cause under-optimization.

4.5.5 Ablation on Nighttime and Rainy Conditions

Because ViLo distills 2D VLM image priors into the 3D network during training, it relies solely on point clouds during inference. To verify its effectiveness under extreme conditions absent in Oxford and NCLT, we conduct experiments on the Boreas dataset [53]. As illustrated in Fig. 6, ViLo is trained on Boreas dataset sequences (A, B and C) and tested across varying conditions (D, E and F), including nighttime (E) and rainy (F) environments. As shown in Tab. 8, ViLo achieves comparable localization performance in both Night (2.63m, 2.48°) and Rainy (2.71m, 2.46°) conditions to that in Sunny (2.33m, 2.68°). This demonstrates the ViLo’s robust environmental adaptability and validates the feasibility of leveraging VLM priors via MOE and distillation. Details in **Appendix Sec. D**.

4.5.6 Visualization of Mixture-of-Experts(MOE) Contributions

Fig. 7 shows MoE contributions across four Oxford regions (A–D), highlighting adaptive routing guided by VLM priors. In occluded narrow streets (A), the model relies on mid- and global-scale experts (Expert- l , g). In vegetation-heavy areas (B), Expert- m dominates to handle uneven density. On open roads with few local features (C), Expert- g ensures stability via global context. At complex intersections (D), expert weights balance, with increased Expert- l for fine details. Overall, ViLo dynamically allocates experts, mitigating performance degradation from capacity interference in a single network across diverse scenes. We also discuss feature distillation in **Appendix Sec. E**.

5 conclusion

We present ViLo, the first LiDAR localization framework to integrate Vision-Language Model (VLM) priors into Scene Coordinate Regression. Guided by these priors, our Scene-Adaptive Mixture-of-Experts (MoE) effectively overcomes capacity interference in degraded scenes. Through cross-modal distillation, ViLo achieves lightweight, LiDAR-only inference without runtime overhead, establishing a new state-of-the-art on the Oxford and NCLT benchmarks.

References

- [1] X. Chen, T. Labe, A. Milioto, T. Rohling, J. Behley, and C. Stachniss, “Overlapnet: A siamese network for computing lidar scan similarity with applications to loop closing and localization,” *Autonomous Robots*, vol. 46, no. 1, pp. 61–81, 2022.
- [2] C. Shi, X. Chen, J. Xiao, B. Dai, and H. Lu, “Fast and accurate deep loop closing and relocalization for reliable lidar slam,” *IEEE Transactions on Robotics*, vol. 40, pp. 2620–2640, 2024.
- [3] H. Yin, X. Xu, S. Lu, X. Chen, R. Xiong, S. Shen, C. Stachniss, and Y. Wang, “A survey on global lidar localization: Challenges, advances and open problems,” *International Journal of Computer Vision*, vol. 132, no. 8, pp. 3139–3171, 2024.
- [4] T. Zhang, G. Wang, Y. Chen, H. Zhang, and J. Hu, “Multi-constellation-inspired single-shot global lidar localization,” in *Proceedings of the AAAI conference on artificial intelligence*, vol. 38, no. 9, 2024, pp. 10 404–10 412.
- [5] J. Komorowski, “Minkloc3d: Point cloud based large-scale place recognition,” in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2021, pp. 1790–1799.
- [6] D. Cattaneo, M. Vaghi, and A. Valada, “Lcdnet: Deep loop closure detection and point cloud registration for lidar slam,” *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2074–2093, 2022.
- [7] K. Xiong, H. Xiang, Q. Xu, C. Wen, S. Shen, J. Li, and C. Wang, “Mining and transferring feature-geometry coherence for unsupervised point cloud registration,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 35 468–35 491, 2024.
- [8] K. Du, H. Xu, H. Li, H. Qu, C.-W. Fu, and S. Liu, “Hybridreg: Robust 3d point cloud registration with hybrid motions,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 3, 2025, pp. 2789–2797.
- [9] Q. Zhou, S. Agostinho, A. Osep, and L. Leal-Taixe, “Is geometry enough for matching in visual localization?” in *European Conference on Computer Vision*. Springer, 2022, pp. 407–425.
- [10] W. Wang, B. Wang, P. Zhao, C. Chen, R. Clark, B. Yang, A. Markham, and N. Trigoni, “Pointloc: Deep pose regressor for lidar point cloud localization,” *IEEE Sensors Journal*, vol. 22, no. 1, pp. 959–968, 2021.
- [11] S. Yu, C. Wang, C. Wen, M. Cheng, M. Liu, Z. Zhang, and X. Li, “Lidar-based localization using universal encoding and memory-aware regression,” *Pattern Recognition*, vol. 128, p. 108685, 2022.
- [12] W. Li, S. Yu, C. Wang, G. Hu, S. Shen, and C. Wen, “Sgloc: Scene geometry encoding for outdoor lidar localization,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 9286–9295.
- [13] S. Wang, Q. Kang, R. She, W. Wang, K. Zhao, Y. Song, and W. P. Tay, “Hypilloc: Towards effective lidar pose regression with hyperbolic fusion,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 5176–5185.
- [14] S. Yu, C. Wang, Y. Lin, C. Wen, M. Cheng, and G. Hu, “Stcloc: Deep lidar localization with spatio-temporal constraints,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 1, pp. 489–500, 2022.
- [15] W. Li, Y. Yang, S. Yu, G. Hu, C. Wen, M. Cheng, and C. Wang, “Diffloc: Diffusion model for outdoor lidar localization,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 15 045–15 054.
- [16] Y. Chen, Q. Li, Y. Yang, W. Li, S. Ao, and C. Wang, “Unleashing the power of data generation in one-pass outdoor lidar localization,” in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 199–208.
- [17] O. Chum and J. Matas, “Optimal randomized ransac,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 30, no. 8, pp. 1472–1482, 2008.

- [18] Y. Yang, W. Li, S. Ao, Q. Xu, S. Yu, Y. Guo, Y. Zhou, S. Shen, and C. Wang, “Raloc: Enhancing outdoor lidar localization via rotation awareness,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 3304–3313.
- [19] B. Yang, Z. Li, W. Li, Z. Cai, C. Wen, Y. Zang, M. Muller, and C. Wang, “Lisa: Lidar localization with semantic awareness,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 15 271–15 280.
- [20] S. Yu, W. Li, X. Sun, Z. Yuan, X. Wang, S. Wang, R. She, and C. Wang, “Gtr-loc: Geospatial text regularization assisted outdoor lidar localization,” in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.
- [21] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.
- [22] W. Li, C. Liu, S. Yu, D. Liu, Y. Zhou, S. Shen, C. Wen, and C. Wang, “Lightloc: Learning outdoor lidar localization at light speed,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 6680–6689.
- [23] W. Li, D. Wang, Z. Ding, A. Sohrabizadeh, Z. Qin, J. Cong, and Y. Sun, “Hierarchical mixture of experts: Generalizable learning for high-level synthesis,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 17, 2025, pp. 18 476–18 484.
- [24] P. Wang, Z. Fan, T. Chen, and Z. Wang, “Neural implicit dictionary learning via mixture-of-expert training,” in *International Conference on Machine Learning*. PMLR, 2022, pp. 22 613–22 624.
- [25] Y. Zhang, R. Cai, T. Chen, G. Zhang, H. Zhang, P.-Y. Chen, S. Chang, Z. Wang, and S. Liu, “Robust mixture-of-expert training for convolutional neural networks,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 90–101.
- [26] J. Wu, X. Li, S. Xu, H. Yuan, H. Ding, Y. Yang, X. Li, J. Zhang, Y. Tong, X. Jiang *et al.*, “Towards open vocabulary learning: A survey,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 46, no. 7, pp. 5092–5113, 2024.
- [27] S. Gupta, J. Hoffman, and J. Malik, “Cross modal distillation for supervision transfer,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2827–2836.
- [28] J. Gou, B. Yu, S. J. Maybank, and D. Tao, “Knowledge distillation: A survey,” *International journal of computer vision*, vol. 129, no. 6, pp. 1789–1819, 2021.
- [29] D. Barnes, M. Gadd, P. Murcutt, P. Newman, and I. Posner, “The oxford radar robotcar dataset: A radar extension to the oxford robotcar dataset,” in *2020 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2020, pp. 6433–6438.
- [30] N. Carlevaris-Bianco, A. K. Ushani, and R. M. Eustice, “University of michigan north campus long-term vision and lidar dataset,” *The International Journal of Robotics Research*, vol. 35, no. 9, pp. 1023–1035, 2016.
- [31] L. Luo, S. Zheng, Y. Li, Y. Fan, B. Yu, S.-Y. Cao, J. Li, and H.-L. Shen, “Bevplace: Learning lidar-based place recognition using bird’s eye view images,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 8700–8709.
- [32] L. Luo, S.-Y. Cao, X. Li, J. Xu, R. Ai, Z. Yu, and X. Chen, “Bevplace++: Fast, robust, and lightweight lidar global localization for unmanned ground vehicles,” *IEEE Transactions on Robotics*, 2025.
- [33] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, “Pointnet++: Deep hierarchical feature learning on point sets in a metric space,” *Advances in neural information processing systems*, vol. 30, 2017.
- [34] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray *et al.*, “Training language models to follow instructions with human feedback,” *Advances in neural information processing systems*, vol. 35, pp. 27 730–27 744, 2022.

- [35] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar *et al.*, “Llama: Open and efficient foundation language models. arXiv 2023,” *arXiv preprint arXiv:2302.13971*, vol. 10, 2023.
- [36] G. Comanici, E. Bieber, M. Schaekermann, I. Pasupat, N. Sachdeva, I. Dhillon, M. Blistein, O. Ram, D. Zhang, E. Rosen *et al.*, “Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities,” *arXiv preprint arXiv:2507.06261*, 2025.
- [37] Y. Qiao, Z. Yu, L. Guo, S. Chen, Z. Zhao, M. Sun, Q. Wu, and J. Liu, “Vl-mamba: Exploring state space models for multimodal learning,” *arXiv preprint arXiv:2403.13600*, 2024.
- [38] S. Kang, Y. Liao, P. Wang, W. Liao, Q. Zhang, B. Busam, X. Chen, and Y. Liu, “Vlm-loc: Localization in point cloud maps via vision-language models,” *arXiv preprint arXiv:2603.09826*, 2026.
- [39] W. Zhang, Y. Guo, L. Niu, P. Li, C. Zhang, Z. Wan, J. Yan, F. U. D. Farrukh, and D. Zhang, “Lp-slam: Language-perceptive rgb-d slam system based on large language model,” *arXiv preprint arXiv:2303.10089*, 2023.
- [40] B. Li, D. Zou, Y. Huang, X. Niu, L. Pei, and W. Yu, “Textslam: Visual slam with semantic planar text features,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 1, pp. 593–610, 2023.
- [41] M. Kolmet, Q. Zhou, A. Ošep, and L. Leal-Taixé, “Text2pos: Text-to-point-cloud cross-modal localization,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 6687–6696.
- [42] Y. Xia, L. Shi, Z. Ding, J. F. Henriques, and D. Cremers, “Text2loc: 3d point cloud localization from natural language,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 14 958–14 967.
- [43] X. Huang, S. Zhao, Y. Wang, X. Lu, W. Zhang, R. Qu, W. Li, Y. Wang, and C. Wen, “Msgnav: Unleashing the power of multi-modal 3d scene graph for zero-shot embodied navigation,” *arXiv preprint arXiv:2511.10376*, 2025.
- [44] Q. Zhao, Y. Lu, M. J. Kim, Z. Fu, Z. Zhang, Y. Wu, Z. Li, Q. Ma, S. Han, C. Finn *et al.*, “Cot-vla: Visual chain-of-thought reasoning for vision-language-action models,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 1702–1713.
- [45] X. Zhao, C. Liu, R. Ji, Z. Zhang, M. Zhu, L. Song, Z. Ren, L. Qingliang, Y. Gao, Z. Du *et al.*, “Cot-vlnbench: A benchmark for visual chain-of-thought reasoning in vision-language-navigation robots,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 43, 2026, pp. 36 573–36 581.
- [46] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *arXiv preprint arXiv:1412.6980*, 2014.
- [47] Y. Zhang, M. Li, D. Long, X. Zhang, H. Lin, B. Yang, P. Xie, A. Yang, D. Liu, J. Lin *et al.*, “Qwen3 embedding: Advancing text embedding and reranking through foundation models,” *arXiv preprint arXiv:2506.05176*, 2025.
- [48] A. Hurst, A. Lerer, A. P. Goucher, A. Perelman, A. Ramesh, A. Clark, A. Ostrow, A. Welihinda, A. Hayes, A. Radford *et al.*, “Gpt-4o system card,” *arXiv preprint arXiv:2410.21276*, 2024.
- [49] B. Hui, J. Yang, Z. Cui, J. Yang, D. Liu, L. Zhang, T. Liu, J. Zhang, B. Yu, K. Lu *et al.*, “Qwen2.5-coder technical report,” *arXiv preprint arXiv:2409.12186*, 2024.
- [50] H. S. Vera, S. Dua, B. Zhang, D. Salz, R. Mullins, S. R. Panyam, S. Smoot, I. Naim, J. Zou, F. Chen *et al.*, “Embeddinggemma: Powerful and lightweight text representations,” *arXiv preprint arXiv:2509.20354*, 2025.
- [51] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” *arXiv preprint arXiv:2402.03216*, vol. 4, no. 5, 2024.

- [52] Z. Nussbaum, J. X. Morris, B. Duderstadt, and A. Mulyar, “Nomic embed: Training a reproducible long context text embedder,” *arXiv preprint arXiv:2402.01613*, 2024.
- [53] K. Burnett, D. J. Yoon, Y. Wu, A. Z. Li, H. Zhang, S. Lu, J. Qian, W.-K. Tseng, A. Lambert, K. Y. Leung *et al.*, “Boreas: A multi-season autonomous driving dataset,” *The International Journal of Robotics Research*, vol. 42, no. 1-2, pp. 33–42, 2023.
- [54] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby *et al.*, “Dinov2: Learning robust visual features without supervision,” *arXiv preprint arXiv:2304.07193*, 2023.